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EDUCATION

UNIVERSITY OF CALIFORNIA AT BERKELEY — DEC 2005
Ph.D. in Mechanical Engineering
Dissertation: Flame Structure and Soot Formation in Inverse Diffusion Flames
Major: Combustion. *Minors:* Fluid Mechanics and Air Quality
Advisor: Professor A.C. Fernandez-Pello

UNIVERSITY OF CALIFORNIA AT BERKELEY — MAY 2004
M.S. in Mechanical Engineering

BOSTON UNIVERSITY — MAY 1993
B.S. in Mechanical Engineering, *Cum Laude*

PROFESSIONAL EXPERIENCE

DIRECTOR OF SOLAR ENERGY ASSESSMENT — 2023 – PRESENT
DNV, Oakland, CA
Lead a team of 10 solar energy analysts evaluating over 500 utility scale & distributed generation projects totaling over 50-GW, generating over $5M. Meet with developers, off-takers, investors, engineering, procurement, & construction companies, create & execute roadmaps to reduce P50 bias & increase confidence.

TEAM LEAD OF SPECIAL PROJECTS FOR SOLAR ENERGY ASSESSMENT — 2021 – 2023
DNV, Oakland CA
Evaluate PV system performance and uncertainty for developers and investors. Create Python libraries to develop, standardize, and accelerate new analyses like solar + storage. Collaborate with national labs and industry partners on methods to evaluate new technology like terrain, bifacial, and hourly modeling corrections.

PRODUCT MANAGER & MODELING SPECIALIST FOR SOLARFARMER — 2017 – 2021
DNV, Oakland CA
Lead product development of disruptive PV-system energy performance software delivering accurate energy assessment coupled with 3-D layout for optimization on any terrain. Implement and validate algorithms. Present at conferences and publish in journals.

SENIOR STAFF RELIABILITY/PERFORMANCE ENGINEER — 2010 – 2017
SunPower Corp., Richmond CA
Develop models to predict energy and degradation of PV systems; conduct research on irradiance, cell, module and inverter performance; design software applications on various platforms, setup servers, databases, CI and SCM repositories, manage and organize projects and co-workers, present at conferences.

SENIOR RESEARCHER, MODELING & PERFORMANCE — 2007 – 2010
AREVA Solar (formerly Ausra), Mountain View CA
Developed and validated dynamic multi-physics models for design and operation of direct steam solar thermal power system. Modeled solar thermal concentrating optics, heat losses, two-phase thermodynamics and fluid mechanics, and thermal storage systems. Analyzed performance tests.

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PUBLICATIONS & CONFERENCE PRESENTATIONS

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PUBLICATIONS

- Jensen A.R., Anderson K.S., Holmgren W.F., Mikofski M.A., Hansen C.W., Boeman L.J., Loonen R., "**pvlib iotools—Open-source Python functions for seamless access to solar irradiance data**," (2023) Solar Energy, 266, DOI: 10.1016/j.solener.2023.112092.
- Deceglie M.G., Anderson K., Fregosi D., Hobbs W.B., Mikofski M.A., Theristis M., Meyers B.E., "**Perspective: Performance Loss Rate in Photovoltaic Systems**," (2023) Solar RRL, 7 (15), DOI: 10.1002/solr.202300196.
- Anderson et al., (2023). "**pvlib python: 2023 project update**." Journal of Open Source Software, 8(92), 5994, DOI: 10.21105/joss.05994.
- M. A. Mikofski, W. F. Holmgren, J. Newmiller and R. Kharait, "**Effects of Solar Resource Sampling Rate and Averaging Interval on Hourly Modeling Errors**," in *IEEE Journal of Photovoltaics*, vol. 13, no. 2, pp. 202-207, March 2023, DOI: 10.1109/JPHOTOV.2023.3238512.
- M. Leung et al., "**Tracker Terrain Loss Part Two**," in *IEEE Journal of Photovoltaics*, vol. 12, no. 1, pp. 127-132, Jan. 2022, DOI: 10.1109/JPHOTOV.2021.3114599.
- G. M. Wilson *et al.*, "**The 2020 photovoltaic technologies roadmap**," *J. Phys. D. Appl. Phys.*, vol. 53, no. 49, 2020. "Ch. 10: Energy Yield Modeling" with M. Deceglie and J. Freeman, DOI: 10.1088/1361-6463/ab9c6a
- K. Anderson and M. A. Mikofski, "**Slope-Aware Backtracking for Single-Axis Trackers**," *National Renewable Energy Laboratory (NREL) Report*, NREL/TP-5K00-76626, Jul. 2020, DOI: 10.2172/1660126.
- W. F. Holmgren, C. W. Hansen, and M. A. Mikofski, "**pvlib python: a python package for modeling solar energy systems**," *J. Open Source Softw.*, vol. 3, no. 29, p. 884, Sep. 2018, DOI: 10.21105/joss.00884

CONFERENCE PRESENTATIONS & POSTERS

- J. Newmiller, M. Mikofski and W. F. Holmgren, "**Photovoltaic Power Plant Capacity Test Uncertainty**," 2025 IEEE 53rd Photovoltaic Specialists Conference (PVSC), Montreal, QC, Canada, 2025, pp. 1347-1352, DOI: 10.1109/PVSC59419.2025.11133042.
- A. Quattrone, A. Pierce, U. Zandanbal and M. A. Mikofski, "**Validation of a hybrid model combining Townsend and Marion to estimate PV snow soiling losses**," 2025 IEEE 53rd Photovoltaic Specialists Conference (PVSC), Montreal, QC, Canada, 2025, pp. 1532-1533, DOI: 10.1109/PVSC59419.2025.11133135.
- U. Akkoseoglu, W. F. Holmgren, M. Mikofski and J. Newmiller, "**Spectral Correction Variability and Energy Yield Impact for cSi Systems and the Effect of Clouds**," 2025 IEEE 53rd Photovoltaic Specialists Conference (PVSC), Montreal, QC, Canada, 2025, pp. 0836-0840, DOI: 10.1109/PVSC59419.2025.11133250.
- K. S. Anderson, C. W. Hansen, W. F. Holmgren, A. R. Jensen, M. A. Mikofski and A. Driesse, "**Pvlib Python: Recent Updates and Community Growth**," 2024 IEEE 52nd Photovoltaic Specialist Conference (PVSC), Seattle, WA, USA, 2024, pp. 1384-1386, DOI: 10.1109/PVSC57443.2024.10749197.
- U. Akkoseoglu, W. F. Holmgren, M. Mikofski and I. Tse, "**Spectral Correction Method Comparison Between First Solar Modeling Tool and PVSyst First Solar Spectral Correction Models**," 2024 IEEE 52nd Photovoltaic Specialist Conference (PVSC), Seattle, WA, USA, 2024, pp. 1105-1108, DOI: 10.1109/PVSC57443.2024.10748936.
- J. Newmiller, M. Mikofski and W. F. Holmgren, "**Bifacial ASTM2848 Capacity Testing with Outboard Sensors**," 2024 IEEE 52nd Photovoltaic Specialist Conference (PVSC), Seattle, WA, USA, 2024, pp. 0867-0870, DOI: 10.1109/PVSC57443.2024.10749204.

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- P. N. Krishnani, A. E. Hajj, C. Helms, S. Nagarajan and M. Mikofski, **"Plane of Array Irradiance Cleaning and Generation of Validated POA Readings for Plant Evaluation,"** 2023 IEEE 50th Photovoltaic Specialists Conference (PVSC), San Juan, PR, USA, 2023, pp. 1-4, DOI: 10.1109/PVSC48320.2023.10359716.
- C. Hayes, A. Parikh, M. Mikofski and R. Kharait, **"Sensitivity of Sub-Hourly Modeling Error to Project Size,"** 2022 IEEE 49th Photovoltaics Specialists Conference (PVSC), Philadelphia, PA, USA, 2022, pp. 0858-0858, DOI: 10.1109/PVSC48317.2022.9938848.
- M. A. Mikofski, W. F. Holmgren, J. Newmiller and R. Kharait, **"Effects of Satellite Sampling on Subhourly Modeling Errors,"** 2022 IEEE 49th Photovoltaics Specialists Conference (PVSC), Philadelphia, PA, USA, 2022, pp. 0995-0995, doi: 10.1109/PVSC48317.2022.9938780.
- K. S. Anderson, W. B. Hobbs, W. F. Holmgren, K. R. Perry, M. A. Mikofski and R. A. Kharait, "**The Effect of Inverter Loading Ratio on Energy Estimate Bias**," *2022 IEEE 49th Photovoltaics Specialists Conference (PVSC)*, 2022, pp. 0714-0720, DOI: 10.1109/PVSC48317.2022.9938632.
- A. Parikh, K. Perry, K. Anderson, W. B. Hobbs, R. Kharait and M. A. Mikofski, "**Validation of Subhourly Clipping Loss Error Corrections**," *2021 IEEE 48th Photovoltaic Specialists Conference (PVSC)*, 2021, pp. 1670-1675, DOI: 10.1109/PVSC43889.2021.9518564.
- M. A. Mikofski and R. Kharait, "**Comparison of Predicted PV System Performance with SURFRAD versus TMY**," *2021 IEEE 48th Photovoltaic Specialists Conference (PVSC)*, 2021, pp. 2155-2159, DOI: 10.1109/PVSC43889.2021.9519024.
- A. Neubert, M. Hamer, R. A. Kharait and M. A. Mikofski, "**Bifacial Solar Sensitivity to Project Capacity Size**," *2020 47th IEEE Photovoltaic Specialists Conference (PVSC)*, 2020, pp. 0703-0706, DOI: 10.1109/PVSC45281.2020.9300712.
- E. L. Warren *et al.*, "**The value of diversity in the renewable energy industry and research community**," *2020 47th IEEE Photovoltaic Specialists Conference (PVSC)*, 2020, pp. 0652-0654, DOI: 10.1109/PVSC45281.2020.9300582.
- R. Kharait, S. Raju, A. Parikh, M. A. Mikofski and J. Newmiller, "**Energy Yield and Clipping Loss Corrections for Hourly Inputs in Climates with Solar Variability**," *2020 47th IEEE Photovoltaic Specialists Conference (PVSC)*, 2020, pp. 1330-1334, DOI: 10.1109/PVSC45281.2020.9300911.
- M. A. Mikofski and P. J. Rainey, "**Tracker Terrain Losses**," *2020 47th IEEE Photovoltaic Specialists Conference (PVSC)*, 2020, pp. 1859-1862, DOI: 10.1109/PVSC45281.2020.9300381.
- M. A. Mikofski, R. Darawali, M. Hamer, A. Neubert, and J. Newmiller, "**Bifacial Performance Modeling in Large Arrays**," in *2019 IEEE 46th Photovoltaic Specialist Conference (PVSC)*, 2019, DOI: 10.1109/PVSC40753.2019.8980572
- W. F. Holmgren, C. W. Hansen, J. S. Stein, and M. A. Mikofski, "**Review of Open Source Tools for PV Modeling**," in *2018 IEEE 7th World Conference on Photovoltaic Energy Conversion (WCPEC) (A Joint Conference of 45th IEEE PVSC, 28th PVSEC & 34th EU PVSEC)*, 2018, pp. 2557–2560, DOI: 10.1109/PVSC.2018.8548231
- M. A. Mikofski, *et al.*, "**Accurate Performance Predictions of Large PV Systems with Shading using Submodule Mismatch Calculation**," in *2018 IEEE 7th World Conference on Photovoltaic Energy Conversion (WCPEC) (A Joint Conference of 45th IEEE PVSC, 28th PVSEC & 34th EU PVSEC)*, 2018, pp. 3635–3639, DOI: 10.1109/PVSC.2018.8547323
- M. A. Mikofski, C. W. Hansen, W. F. Holmgren, and G. M. Kimball, "**Use of measured aerosol optical depth and precipitable water to model clear sky irradiance**," in *2017 IEEE 44th Photovoltaic Specialist Conference (PVSC)*, 2017, no. July, pp. 110–116, DOI: 10.1109/PVSC.2017.8366314
- B. Meyers and M. Mikofski, "**Accurate Modeling of Partially Shaded PV Arrays**," in *2017 IEEE 44th Photovoltaic Specialist Conference (PVSC)*, 2017, pp. 3354–3359, DOI: 10.1109/PVSC.2017.8521559

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- M. Mikofski, A. Oumbe, C. Li, and B. Bourne, "**Evaluation and correction of the impact of spectral variation of irradiance on PV performance**," in *2016 IEEE 43rd Photovoltaic Specialists Conference (PVSC)*, 2016, pp. 1357–1362, DOI: 10.1109/PVSC.2016.7749837
- B. Meyers, M. Mikofski, and M. Anderson, "**A fast parameterized model for predicting PV system performance under partial shade conditions**," in *2016 IEEE 43rd Photovoltaic Specialists Conference (PVSC)*, 2016, pp. 3173–3178, DOI: 10.1109/PVSC.2016.7750251
- E. Hasselbrink *et al.*, "**Validation of the PVLife model using 3 million module-years of live site data**," in *2013 IEEE 39th Photovoltaic Specialists Conference (PVSC)*, 2013, pp. 7–12. DOI: 10.1109/PVSC.2013.6744087
- M. A. Mikofski *et al.*, "**PVLife: An Integrated Model for Predicting PV Performance Degradation over 25+ Years**," in *2012 38th IEEE Photovoltaic Specialists Conference*, 2012, no. 3, pp. 1744–1749, DOI: 10.1109/PVSC.2012.6317932
- M. Mikofski *et al.*, "**A Dynamic Cell-By-Cell PV System Model to Predict Lifetime Performance and Reliability**," in *26th European Photovoltaic Solar Energy Conference and Exhibition*, 2011, pp. 105–112, DOI: 10.4229/26thEUPVSEC2011-1BO.10.1